

SECTION A: OBJECTIVE TYPE QUESTIONS

Q1. Answer any 4 out of the given 6 questions (1 x 4 = 4 marks)

1. _____ is the final component in the process of communication as it defines the response given by the receiver to the sender.
 - Answer: c) Feedback
 2. _____ refers to focusing human efforts for maintaining a healthy body and mind capable of better withstanding stressful situations.
 - Answer: d) Stress Management
 3. Having conscious knowledge of your own self, capabilities, feelings and one's own character is called _____.
 - Answer: a) Self-awareness
 4. A _____ is a software program that attaches itself to other programs and alters their behavior.
 - Answer: d) Computer Virus
 5. _____ refers to recruitment, employment, selection, training, development and compensation of the employees with an organization.
 - Answer: c) Human Resource Management
 6. _____ is caused when natural or a man-made disturbance disrupts the natural balance of an ecosystem.
 - Answer: d) Ecological Imbalance
-

Q2. Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)

1. A _____ is divided into multiple layers and each layer is further divided into several blocks called nodes.
 - Answer: a) Neural Networks
2. The _____ canvas helps you in identifying the key elements related to the problem.
 - Answer: a) Problem scoping

3. _____ is a domain of AI that depicts the capability of a machine to get and analyze visual information and afterwards predict some decisions about it.

- Answer: d) Computer Vision

4. _____ is defined as the percentage of correct predictions out of all the observations.

- Answer: b) Accuracy

5. _____ is the sub-field of AI that is focused on enabling computers to understand and process human languages.

- Answer: c) NLP

6. In _____, the machine is trained with huge amounts of data which helps it in training itself around the data.

- Answer: b) Deep Learning

Q3. Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)

1. Expand CBT _____

- Answer: b) Cognitive Behavioral Therapy

2. Name any 2 methods of collecting data.

- Answer: a) Surveys and Interviews

3. What is the role of modelling in an NLP based AI model?

- Answer: a) Modelling in NLP helps in processing of AI model

4. What will be the outcome, if the Prediction is “Yes” and it matches with the Reality? What will be the outcome, if the Prediction is “Yes” and it does not match the Reality?

- Answer: d) True Positive, False Positive

5. Recall-Evaluation method is

- Answer: a) defined as the fraction of positive cases that are correctly identified.

6. Give 2 examples of Supervised Learning models.

- Answer: a) Classification and Regression
-

Q4. Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**1. Define Machine Learning.**

- **Answer:** a) Machine learning is the study of computer algorithms that improve automatically through experience.

2. Write common terminologies used in Data.

- **Answer:** Examples include: Data set, Feature, Label, Training Set, Test Set, Model, Algorithm, Accuracy, etc.

3. Differentiate between Prediction and Reality.

- **Answer:** c) The prediction is the output which is given by the machine and the reality is the real scenario in which the prediction has been made.

4. Which of the following best represents a key ethical concern in the development and deployment of AI systems?

- **Answer:** c) Ensuring transparency and accountability in AI decision-making

5. Which of the following statements is true for the term Evaluation?

- **Answer:** c) Helps in understanding the reliability of any AI model

6. Which of the following is not part of the AI Project Cycle?

- **Answer:** d) Problem Scoping
-

Q5. Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)**1. _____ refers to the AI modelling where the machine learns by itself.**

- **Answer:** c) Machine Learning

2. Prediction and Reality can be easily mapped together with the help of:

- **Answer:** d) Confusion Matrix

3. _____ is an example of Applications of Natural Language Processing.

- **Answer:** b) Automatic Summarization

4. _____ is the last stage of the AI project Life cycle.

- **Answer:** b) Evaluation

5. In _____, the machine is trained with huge amounts of data which helps it in training itself around the data.

- **Answer:** d) Deep Learning

6. In _____, input to machines can be photographs, videos and pictures from thermal or infrared sensors, indicators and different sources.

- **Answer:** a) Computer Vision
-

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions (2 x 3 = 6 marks)

1. Name the four main categories of Communication Styles

- **Answer:** The four main categories of communication styles are:

1. Assertive
2. Aggressive
3. Passive
4. Passive-Aggressive

2. List any 4 activities that help in stress management.

- **Answer:**

1. Exercise (e.g., yoga, walking)
2. Breathing exercises
3. Mindfulness meditation
4. Time management

3. What are antivirus? Name any 2 antiviruses.

- **Answer:** Antivirus software is designed to detect, prevent, and remove malicious software (malware) from computers. Examples of antiviruses include:

1. Norton Antivirus
2. McAfee Antivirus

4. Name any 4 qualities of an entrepreneur.

- **Answer:**

1. Innovative
 2. Risk-taker
 3. Leadership skills
 4. Determined
5. Name any 4 man-made disruptions that cause ecological imbalance (Climate change).
- Answer:
 1. Deforestation
 2. Industrial Pollution
 3. Overfishing
 4. Greenhouse gas emissions
-

Answer any 4 out of the given 6 questions in 20–30 words (2 x 4 = 8 marks)

1. Give 2 points of difference between a Script-bot and a Smart-bot
 - Answer:
 - **Script-bot:** Executes pre-defined tasks based on scripts; limited to specific functions.
 - **Smart-bot:** Uses AI to learn and adapt to user behavior and improves performance over time.
2. Define the term Machine Learning. Also, give 2 applications of Machine Learning in our daily lives.
 - Answer:
 - **Machine Learning:** Machine learning refers to algorithms that enable computers to learn patterns from data and improve their performance without explicit programming.
 - **Applications:**
 1. Recommendation Systems (e.g., Netflix, Amazon)
 2. Voice Assistants (e.g., Siri, Alexa)
3. Differentiate between Classification and Regression.
 - Answer:

- **Classification:** Predicts categorical outcomes (e.g., spam or not spam).
- **Regression:** Predicts continuous values (e.g., house prices).

4. Explain various stages of AI Project Cycle

- **Answer:** The AI project cycle includes:
 - **Problem Scoping** – Identifying the problem to solve.
 - **Data Acquisition** – Collecting relevant data.
 - **Data Exploration** – Exploring and cleaning the data.
 - **Modelling** – Building the machine learning model.
 - **Evaluation** – Assessing model performance.
 - **Deployment** – Implementing the model in real-world use.

5. Explain “Bio-Ethics”

- **Answer:** Bioethics is the study of ethical issues emerging from advancements in biology, medicine, and technology, especially as they pertain to human life, health, and the environment.

6. Draw a confusion matrix table and plot Actual vs Prediction and TP, TN, FP, FN

- **Answer:** Here is a basic confusion matrix:

	Predicted: Yes	Predicted: No
Actual: Yes	TP (True Positive)	FN (False Negative)
Actual: No	FP (False Positive)	TN (True Negative)

Answer any 3 out of the given 5 questions in 50–80 words each (4 x 3 = 12 marks)

1. Create a 4W Project Canvas for the following:

Scenario: As more and more new technologies get into play, risks will get more concentrated into a common network. Cybersecurity becomes extremely complicated in such scenarios and goes beyond the control of firewalls. AI algorithms can help scrape through vast amounts of logs to identify susceptible user behavior.

4W Project Canvas:

- **What:** Build an AI-based system to detect unusual activity, risks, and threats in a network through log analysis and behavior prediction.
 - **Why:** To enhance cybersecurity and reduce the risk of data breaches or system vulnerabilities that current firewalls cannot detect.
 - **Who:** Cybersecurity teams, data scientists, and AI engineers.
 - **When:** During real-time network monitoring and in proactive threat detection.
-

2. Write the applications of Data Science (any four)

Applications of Data Science:

1. **Healthcare:** Predicting patient outcomes, diagnosing diseases, and personalizing treatment plans.
2. **Finance:** Fraud detection, risk management, and algorithmic trading.
3. **Marketing:** Customer segmentation, targeted advertising, and sales forecasting.
4. **Retail:** Inventory management, supply chain optimization, and recommendation systems.

These applications demonstrate how Data Science helps various industries make data-driven decisions and optimize their operations.

3. Explain Modelling and types

Modelling in AI and Machine Learning refers to the process of creating a mathematical representation (model) of a real-world problem using data. The model is trained on data and tested for accuracy to predict outcomes or make decisions.

Types of Models:

1. **Supervised Learning:** The model is trained with labeled data. Examples include regression and classification models.
2. **Unsupervised Learning:** The model identifies patterns or structures in data without labels. Examples include clustering and dimensionality reduction.
3. **Reinforcement Learning:** The model learns by interacting with the environment and receiving feedback in the form of rewards or penalties.

4. Explain High code, Low code, No code differences and benefits of "No Code"

- **High Code:** Involves traditional programming where developers write extensive code to create applications, providing complete flexibility.
- **Low Code:** Involves minimal coding with visual interfaces, where developers can drag and drop components to build apps.
- **No Code:** A platform where users can build apps without writing any code, often through drag-and-drop functionality, making it accessible to non-technical users.

Benefits of No Code:

- **Faster development:** Reduces the time to build applications.
 - **Accessibility:** Allows non-technical users to create and modify applications.
 - **Cost-effective:** Reduces the need for specialized developers, lowering project costs.
-

5. Confusion Matrix for Traffic Jam Prediction Model

Given the AI model's task to predict traffic jams, here is how the **Confusion Matrix** would look:

	Predicted: Yes (Traffic Jam)	Predicted: No (No Traffic Jam)
Actual: Yes (Actual Traffic Jam)	TP (True Positive)	FN (False Negative)
Actual: No (Actual No Jam)	FP (False Positive)	TN (True Negative)

- **True Positive (TP):** The model correctly predicted a traffic jam when there was one.
- **False Negative (FN):** The model missed predicting a traffic jam when there was one.
- **False Positive (FP):** The model incorrectly predicted a traffic jam when there was none.
- **True Negative (TN):** The model correctly predicted no traffic jam when there wasn't one.

This helps measure the model's effectiveness in predicting traffic jams, which directly affects users' travel plans.